"PAPER<i>{0:D3}" -f [int]# PAPER #38 ■ F</i>UBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

Author: Daniel T. Murphy

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Title: UQFF Buoyancy Proof Variants 7■11: Fermi Acceleration, Cosmic Ray Knee, WHIM Temperature, Press-Schechter Halos, and Star Formation Efficiency

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** 98b2e77dfbc34d27b09f19fa7c460624 **Validator:** `BuoyancyProofVariants.py` ■ All 17 variants operational ? **Variants:** fermi, kne, whim, ps, sfe **Index Slot:** ■1.5 Buoyancy Proofs, $n = [int]# PAPER #38 ■ F_UBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

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Abstract

This paper presents five F*UBii buoyancy proof variants addressing quantum corrections to macroscopic astrophysical processes. Variant 7 (fermi) derives the UQFF buoyancy of Fermi-accelerated particles at astrophysical shock fronts. Variant 8 (kne) applies the framework to the cosmic ray knee at ~3■10■5 eV where the spectral index changes ■ the UQFF predicts this spectral break as a phase transition in the F*UBii landscape. Variant 9 (whim) addresses the Warm-Hot Intergalactic Medium at T ~ 105■107 K containing 40■50% of cosmic baryons. Variant 10 (ps) maps the Press-Schechter dark matter halo mass function to a UQFF buoyancy force landscape. Variant 11 (sfe) quantifies the UQFF buoyancy contribution to star formation efficiency suppression in molecular clouds. Together these form the quantum corrections series, where small-scale quantum physics drives large-scale structure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Variant 7: Fermi Acceleration Buoyancy (fermi)

1.1 Physical Context

First-order Fermi (DSA) acceleration at shock fronts produces power-law particle spectra N(E) ? E?? with s = (r+2)/(r-1) for compression ratio r. The UQFF buoyancy arises because accelerated particles develop a pressure gradient against the surrounding thermal plasma.

Key systems: Tycho SNR (v_shock ~ 4500 km/s), Centaurus A jet shocks, Cygnus A hotspots

1.2 FUBiifermi Equation

```
F_{\rm UBii,fermi} = F_{\rm rel} \cdot \frac{\beta_{\rm shock} \cdot E_p}{E_{\rm LEP}}
\cdot Q_{\rm wave} \cdot \left(\frac{v_{\rm shock}}{c}\right)^2
```

where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

E_p = particle energy (J)

v_shock = shock velocity (m/s)

1.3 (v/c)■ Relativistic Correction

The Fermi buoyancy scales as (v*shock/c)*■ ■ *a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)*■ *= 10?4 relative to the full E*p contribution.

1.4 Example: Centaurus A Jet Hotspot

For Cen A hotspot: ■*shock = 4 (strong shock), E*p = 10?? J (10 GeV proton), v*shock = 0.5c, Q*wave = 1.0:

```
F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
\times (0.5)^2 = 10^{-10} \times 3.28\times10^{10} \times 0.25 = 8.2\times10^{0} = 0.82
\text{ N}
```

The per-particle Fermi buoyancy force of 0.82 N per 10 GeV proton, scaled over the ~106■ protons in the hotspot, gives the collective Fermi acceleration pressure maintaining the Cen A jet head.

---

2. Variant 8: Cosmic Ray Knee Energy Buoyancy (kne)

2.1 Physical Context

The cosmic ray energy spectrum follows E?■■7 power law up to the "knee" at ~3■10■5 eV, where it steepens to E?■■■. The knee marks the maximum energy achievable by Galactic SNR shock acceleration for protons (Z=1). For heavy nuclei, the knee scales as Z ■ E_knee(proton) ■ the "knee composition model".

2.2 FUBiikne Equation

```
F_{\rm UBii,kne} = -F_{\rm rel} \cdot \frac{E_{\rm knee}}{E_{\rm GUT}} \cdot \frac{Z
\cdot e}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \ln\left(\frac{E_{\rm knee}}{E_{\rm
LEP}}\right)
```

where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

E_GUT = GUT energy scale = 1.6■10?5 J (~10■6 GeV)

Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
35.9
```

This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
imes10^{-19})}{\ln(4.8\times10^{-4}/1.22\times10^{-19})} = 26 \times \frac{38.0}{35.9} =
27.5
```

The UQFF predicts the iron knee force is 27.5■ the proton knee force (vs 26■ in the pure rigidity model), a 5.8% enhancement from the quantum logarithmic correction.

---

3. Variant 9: WHIM Temperature Buoyancy (whim)

3.1 Physical Context

The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

```
F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

T_WHIM = WHIM temperature (K)

n_b = baryon number density (m?■)

s_T = Thomson cross-section = 6.652■10?■? m■

r_fil = filament radius (m)

T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

Factor 1: thermal pressure k*B T*WHIM

Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)

Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K, n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m, T*virial = 107 K, Q_wave = 1.0:

```
F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

M_halo = halo mass (kg)

M_P = Planck mass = 2.176■10?8 kg

d_c = 1.686 (critical overdensity)

s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of MP■ *(the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10*■■ *M? = 1.989*■*104*■ *kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M**/M*gas ranges from ~1% in diffuse GMCs to ~30*■*50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
```

This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086*■*10*■*7 m, Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
(3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
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= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 =
1.72\times10^{22} \text{ N}
```

---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
kne	CR knee at 3■10■5 eV	E*knee/E*GUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		

Variant	Physical Context	Key Formula	Characteristic Scale		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		
ps	PS halo mass function	M*halo*/*M*P■ ■ d_c ■	dln s/dln M		~1068 N (MW scale)
sfe	Molecular cloud SFR	e^(3/2) ■ M_gas c■/r■	~10■■ N (Orion A)		

Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
- **kne:** CR knee is the stationary point of F*UBii*kne ■ a UQFF phase transition rather than a diffusive escape threshold
- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
- **sfe:** Star formation efficiency is UQFF-limited by rest-mass energy Bekenstein-area scaling: F*UBii*sfe ? e^(3/2) M_gas c■/r■

Validator: `BuoyancyProofVariants.py` *? All 17 FUBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER*{0:D3}" -f $n

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where:

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Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee*/*E*LEP) factor:

$$\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) = 35.9$$

This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

$$\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm knee}$$

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

$$F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\times10^{-19})}{\ln(4.8\times10^{-4}/1.22\times10^{-19})} = 26 \times \frac{38.0}{35.9} = 27.5$$

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where:

T_WHIM = WHIM temperature (K)

n_b = baryon number density (m?■)

s_T = Thomson cross-section = 6.652■10?■? m■

r_fil = filament radius (m)

T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

- Factor 1: thermal pressure k*B T*WHIM
- Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)
- Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

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For a typical baryon-rich filament: T*WHIM = 106 K*, *n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m*, *T*virial = 107 K, Q_wave = 1.0:

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This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

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The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

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\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
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M}\right)
```

where:

- M_halo = halo mass (kg)
- M_P = Planck mass = 2.176■10?8 kg
- d_c = 1.686 (critical overdensity)
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4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of M*P■ (the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
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```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
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---

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Star formation efficiency e*SFE = M**/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
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```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
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For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m, Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
(3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
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---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
kne	CR knee at 3■10■5 eV	E*knee/E*GUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		

Variant	Physical Context	Key Formula	Characteristic Scale		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		
ps	PS halo mass function	M*halo*/*M*P■ ■ d_c ■	dln s/dln M		~1068 N (MW scale)
sfe	Molecular cloud SFR	e^(3/2) ■ M_gas c■/r■	~10■■ N (Orion A)		

Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
- **kne:** CR knee is the stationary point of F*UBii*kne ■ a UQFF phase transition rather than a diffusive escape threshold
- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
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Validator: `BuoyancyProofVariants.py` *? All 17 FUBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value* ■ *F*UBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

Title: UQFF Buoyancy Proof Variants 7■11: Fermi Acceleration, Cosmic Ray Knee, WHIM Temperature, Press-Schechter Halos, and Star Formation Efficiency

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** 98b2e77dfbc34d27b09f19fa7c460624 **Validator:** `BuoyancyProofVariants.py` ■ All 17 variants operational ? **Variants:** fermi, kne, whim, ps, sfe **Index Slot:** ■1.5 Buoyancy Proofs, $n = [int]# "PAPER*{0:D3}" -f [int]# PAPER #38* ■ *F*UBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

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This paper presents five F*UBii buoyancy proof variants addressing quantum corrections to macroscopic astrophysical processes. Variant 7 (fermi) derives the UQFF buoyancy of Fermi-accelerated particles at astrophysical shock fronts. Variant 8 (kne) applies the framework to the cosmic ray knee at ~3■10■5 eV where the spectral index changes ■ the UQFF predicts this spectral break as a phase transition in the F*UBii landscape. Variant 9 (whim) addresses the Warm-Hot Intergalactic Medium at T ~ 105■107 K containing 40■50% of cosmic baryons. Variant 10 (ps) maps the Press-Schechter dark matter halo mass function to a UQFF buoyancy force landscape. Variant 11 (sfe) quantifies the UQFF buoyancy contribution to star formation efficiency suppression in molecular clouds. Together these form the quantum corrections series, where small-scale quantum physics drives large-scale structure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Variant 7: Fermi Acceleration Buoyancy (fermi)

1.1 Physical Context

First-order Fermi (DSA) acceleration at shock fronts produces power-law particle spectra N(E) ? E?? with s = (r+2)/(r-1) for compression ratio r. The UQFF buoyancy arises because accelerated particles develop a pressure gradient against the surrounding thermal plasma.

Key systems: Tycho SNR (v_shock ~ 4500 km/s), Centaurus A jet shocks, Cygnus A hotspots

1.2 FUBiifermi Equation

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F_{\rm UBii,fermi} = F_{\rm rel} \cdot \frac{\beta_{\rm shock} \cdot E_p}{E_{\rm LEP}}
\cdot Q_{\rm wave} \cdot \left(\frac{v_{\rm shock}}{c}\right)^2
```

where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

E_p = particle energy (J)

v_shock = shock velocity (m/s)

1.3 (v/c)■ Relativistic Correction

The Fermi buoyancy scales as (v*shock/c)■ ■ a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)■ = 10?4 relative to the full E*p contribution.

1.4 Example: Centaurus A Jet Hotspot

For Cen A hotspot: ■*shock = 4 (strong shock), E*p = 10?? J (10 GeV proton), v*shock = 0.5c, Q*wave = 1.0:

```
F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
\times (0.5)^2 = 10^{-10} \times 3.28\times10^{10} \times 0.25 = 8.2\times10^{0} = 0.82
\text{ N}
```

The per-particle Fermi buoyancy force of 0.82 N per 10 GeV proton, scaled over the ~106■ protons in the hotspot, gives the collective Fermi acceleration pressure maintaining the Cen A jet head.

---

2. Variant 8: Cosmic Ray Knee Energy Buoyancy (kne)

2.1 Physical Context

The cosmic ray energy spectrum follows E?■■7 power law up to the "knee" at ~3■10■5 eV, where it steepens to E?■■■. The knee marks the maximum energy achievable by Galactic SNR shock acceleration for protons (Z=1). For heavy nuclei, the knee scales as Z ■ E_knee(proton) ■ the "knee composition model".

2.2 FUBiikne Equation

```
F_{\rm UBii,kne} = -F_{\rm rel} \cdot \frac{E_{\rm knee}}{E_{\rm GUT}} \cdot \frac{Z
\cdot e}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \ln\left(\frac{E_{\rm knee}}{E_{\rm
LEP}}\right)
```

where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

E_GUT = GUT energy scale = 1.6■10?5 J (~10■6 GeV)

Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
35.9
```

This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a **stationary point** of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
imes10^{-19})}{\ln(4.8\times10^{-4}/1.22\times10^{-19})} = 26 \times \frac{38.0}{35.9} =
27.5
```

The UQFF predicts the iron knee force is 27.5■ the proton knee force (vs 26■ in the pure rigidity model), a 5.8% enhancement from the quantum logarithmic correction.

---

3. Variant 9: WHIM Temperature Buoyancy (whim)

3.1 Physical Context

The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

```
F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

T_WHIM = WHIM temperature (K)

n_b = baryon number density (m?■)

s_T = Thomson cross-section = 6.652■10?■? m■

r_fil = filament radius (m)

T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

Factor 1: thermal pressure k*B T*WHIM

Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)

Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K, n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m, T*virial = 107 K, Q_wave = 1.0:

```
F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

M_halo = halo mass (kg)

M_P = Planck mass = 2.176■10?8 kg

d_c = 1.686 (critical overdensity)

s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of M*P■ (the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M*/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum

[SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
```

This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05*, *M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m*, *Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times (3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times 1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 = 1.72\times10^{22} \text{ N}
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---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
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kne	CR knee at 3■10■5 eV	*Eknee/E*GUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		
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Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

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where:

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e = 1.602■10?■? C

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The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

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The ln(E*knee/E*LEP) factor:

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This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
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For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

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The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

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F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

T_WHIM = WHIM temperature (K)

n_b = baryon number density (m?■)

s_T = Thomson cross-section = 6.652■10?■? m■

r_fil = filament radius (m)

T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

Factor 1: thermal pressure k*B T*WHIM

Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)

Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K, n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m, T*virial = 107 K, Q_wave = 1.0:

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F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
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This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

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The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
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```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

M_halo = halo mass (kg)

M_P = Planck mass = 2.176■10?8 kg

d_c = 1.686 (critical overdensity)

s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/MP*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of MP■ *(the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
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```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M**/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum

[SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
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This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05*, *M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m*, *Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times (3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times 1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 = 1.72\times10^{22} \text{ N}
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---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
kne	CR knee at 3■10■5 eV	*Eknee*/*E*GUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		
ps	PS halo mass function	*Mhalo*/*MP*■ ■ d_c ■	dln s/dln M		~1068 N (MW scale)
sfe	Molecular cloud SFR	e^(3/2) ■ M_gas c■/r■	~10■■ N (Orion A)		

Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
- **kne:** CR knee is the stationary point of F*UBii*kne ■ a UQFF phase transition rather than a diffusive escape threshold
- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
- **sfe:** Star formation efficiency is UQFF-limited by rest-mass energy Bekenstein-area scaling: F*UBii*sfe ? e^(3/2) M_gas c■/r■

Validator: `BuoyancyProofVariants.py` *? All 17 FUBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value* ■ FUBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

Title: UQFF Buoyancy Proof Variants 7■11: Fermi Acceleration, Cosmic Ray Knee, WHIM Temperature, Press-Schechter Halos, and Star Formation Efficiency

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** 98b2e77dfbc34d27b09f19fa7c460624 **Validator:** `BuoyancyProofVariants.py` ■ All 17 variants operational ? **Variants:** fermi, kne, whim, ps, sfe **Index Slot:** ■1.5 Buoyancy Proofs, "PAPER*{0:D3}" -f [int]# PAPER #38* ■ *F*UBii Buoyancy Force: Proof Variants 7■11 (Quantum Corrections)

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Abstract

This paper presents five F*UBii buoyancy proof variants addressing quantum corrections to macroscopic astrophysical processes. Variant 7 (fermi) derives the UQFF buoyancy of Fermi-accelerated particles at astrophysical shock fronts. Variant 8 (kne) applies the framework to the cosmic ray knee at ~3■10■5 eV where the spectral index changes ■ the UQFF predicts this spectral break as a phase transition in the F*UBii landscape. Variant 9 (whim) addresses the Warm-Hot Intergalactic Medium at T ~ 105■107 K containing 40■50% of cosmic baryons. Variant 10 (ps) maps the Press-Schechter dark matter halo mass function to a UQFF buoyancy force landscape. Variant 11 (sfe) quantifies the UQFF buoyancy contribution to star formation efficiency suppression in molecular clouds. Together these form the quantum corrections series, where small-scale quantum physics drives large-scale structure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. Variant 7: Fermi Acceleration Buoyancy (fermi)

1.1 Physical Context

First-order Fermi (DSA) acceleration at shock fronts produces power-law particle spectra N(E) ? E?? with s = (r+2)/(r-1) for compression ratio r. The UQFF buoyancy arises because accelerated particles develop a pressure gradient against the surrounding thermal plasma.

Key systems: Tycho SNR (v_shock ~ 4500 km/s), Centaurus A jet shocks, Cygnus A hotspots

1.2 FUBiifermi Equation

```
F_{\rm UBii,fermi} = F_{\rm rel} \cdot \frac{\beta_{\rm shock} \cdot E_p}{E_{\rm LEP}}
\cdot Q_{\rm wave} \cdot \left(\frac{v_{\rm shock}}{c}\right)^2
```

where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

E_p = particle energy (J)

v_shock = shock velocity (m/s)

1.3 (v/c)■ Relativistic Correction

The Fermi buoyancy scales as (v*shock/c)*■ ■ *a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)*■ *= 10?4 relative to the full E*p contribution.

1.4 Example: Centaurus A Jet Hotspot

For Cen A hotspot: ■*shock = 4 (strong shock), E*p = 10?? J (10 GeV proton), v*shock = 0.5c, Q*wave = 1.0:

```
F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
\times (0.5)^2 = 10^{-10} \times 3.28\times10^{10} \times 0.25 = 8.2\times10^{0} = 0.82
\text{ N}
```

The per-particle Fermi buoyancy force of 0.82 N per 10 GeV proton, scaled over the ~106■ protons in the hotspot, gives the collective Fermi acceleration pressure maintaining the Cen A jet head.

---

2. Variant 8: Cosmic Ray Knee Energy Buoyancy (kne)

2.1 Physical Context

The cosmic ray energy spectrum follows E?■■7 power law up to the "knee" at ~3■10■5 eV, where it steepens to E?■■■. The knee marks the maximum energy achievable by Galactic SNR shock acceleration for protons (Z=1). For heavy nuclei, the knee scales as Z ■ E_knee(proton) ■ the "knee composition model".

2.2 FUBiikne Equation

```
F_{\rm UBii,kne} = -F_{\rm rel} \cdot \frac{E_{\rm knee}}{E_{\rm GUT}} \cdot \frac{Z
\cdot e}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \ln\left(\frac{E_{\rm knee}}{E_{\rm
LEP}}\right)
```

where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

E_GUT = GUT energy scale = 1.6■10?5 J (~10■6 GeV)

Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
35.9
```

This logarithm attains its maximum precisely at E*knee for protons* ■ *a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
imes10^{-19})}{\ln(4.8\times10^{-4}/1.22\times10^{-19})} = 26 \times \frac{38.0}{35.9} =
27.5
```

The UQFF predicts the iron knee force is 27.5■ the proton knee force (vs 26■ in the pure rigidity model), a 5.8% enhancement from the quantum logarithmic correction.

---

3. Variant 9: WHIM Temperature Buoyancy (whim)

3.1 Physical Context

The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

```
F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

- T_WHIM = WHIM temperature (K)
- n_b = baryon number density (m?■)
- s_T = Thomson cross-section = 6.652■10?■? m■
- r_fil = filament radius (m)
- T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

- Factor 1: thermal pressure k*B T*WHIM
- Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)

Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K*, *n*b = 10 m?■, r*fil = 10 Mpc = 3.09*■*10*■■ *m*, *T*virial = 107 K, Q_wave = 1.0:

```
F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

M_halo = halo mass (kg)

M_P = Planck mass = 2.176■10?8 kg

d_c = 1.686 (critical overdensity)

s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of M*P*■ *(the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M*/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
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```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
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This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m, Q*wave = 1.0:

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F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
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```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 =
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---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
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kne	CR knee at 3■10■5 eV	E*knee*/EGUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		

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Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

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Validator: `BuoyancyProofVariants.py` *? All 17 FUBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER*{0:D3}" -f $n

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where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

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The Fermi buoyancy scales as (v*shock/c)■ ■ a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)■ = 10?4 relative to the full E*p contribution.

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F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
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where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

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Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
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This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
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3. Variant 9: WHIM Temperature Buoyancy (whim)

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The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

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```

where:

- T_WHIM = WHIM temperature (K)
- n_b = baryon number density (m?■)
- s_T = Thomson cross-section = 6.652■10?■? m■
- r_fil = filament radius (m)
- T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

Factor 1: thermal pressure k*B T*WHIM

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Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K, n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m, T*virial = 107 K, Q_wave = 1.0:

```
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\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

M_halo = halo mass (kg)

M_P = Planck mass = 2.176■10?8 kg

d_c = 1.686 (critical overdensity)

s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of MP■ *(the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M**/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
```

This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m, Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
(3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 =
1.72\times10^{22} \text{ N}
```

---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
kne	CR knee at 3■10■5 eV	E*knee*/EGUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		

Variant	Physical Context	Key Formula	Characteristic Scale		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		
ps	PS halo mass function	M*halo*/*M*P■ ■ d_c ■	dln s/dln M		~1068 N (MW scale)
sfe	Molecular cloud SFR	e^(3/2) ■ M_gas c■/r■	~10■■ N (Orion A)		

Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
- **kne:** CR knee is the stationary point of F*UBii*kne ■ a UQFF phase transition rather than a diffusive escape threshold
- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
- **sfe:** Star formation efficiency is UQFF-limited by rest-mass energy Bekenstein-area scaling: F*UBii*sfe ? e^(3/2) M_gas c■/r■

Validator: `BuoyancyProofVariants.py` *? All 17 F_UBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value

---

Abstract

This paper presents five F*UBii buoyancy proof variants addressing quantum corrections to macroscopic astrophysical processes. Variant 7 (fermi) derives the UQFF buoyancy of Fermi-accelerated particles at astrophysical shock fronts. Variant 8 (kne) applies the framework to the cosmic ray knee at ~3■10■5 eV where the spectral index changes ■ the UQFF predicts this spectral break as a phase transition in the F*UBii landscape. Variant 9 (whim) addresses the Warm-Hot Intergalactic Medium at T ~ 105■107 K containing 40■50% of cosmic baryons. Variant 10 (ps) maps the Press-Schechter dark matter halo mass function to a UQFF buoyancy force landscape. Variant 11 (sfe) quantifies the UQFF buoyancy contribution to star formation efficiency suppression in molecular clouds. Together these form the quantum corrections series, where small-scale quantum physics drives large-scale structure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. Variant 7: Fermi Acceleration Buoyancy (fermi)

1.1 Physical Context

First-order Fermi (DSA) acceleration at shock fronts produces power-law particle spectra N(E) ? E?? with s = (r+2)/(r-1) for compression ratio r. The UQFF buoyancy arises because accelerated particles develop a pressure gradient against the surrounding thermal plasma.

Key systems: Tycho SNR (v_shock ~ 4500 km/s), Centaurus A jet shocks, Cygnus A hotspots

1.2 FUBiifermi Equation

```
F_{\rm UBii,fermi} = F_{\rm rel} \cdot \frac{\beta_{\rm shock} \cdot E_p}{E_{\rm LEP}}
\cdot Q_{\rm wave} \cdot \left(\frac{v_{\rm shock}}{c}\right)^2
```

where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

E_p = particle energy (J)

v_shock = shock velocity (m/s)

1.3 (v/c)■ Relativistic Correction

The Fermi buoyancy scales as (v*shock/c)■ ■ a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)■ = 10?4 relative to the full E*p contribution.

1.4 Example: Centaurus A Jet Hotspot

For Cen A hotspot: ■*shock = 4 (strong shock), E*p = 10?? J (10 GeV proton), v*shock = 0.5c, Q*wave = 1.0:

```
F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
\times (0.5)^2 = 10^{-10} \times 3.28\times10^{10} \times 0.25 = 8.2\times10^{0} = 0.82
\text{ N}
```

The per-particle Fermi buoyancy force of 0.82 N per 10 GeV proton, scaled over the ~106■ protons in the hotspot, gives the collective Fermi acceleration pressure maintaining the Cen A jet head.

---

2. Variant 8: Cosmic Ray Knee Energy Buoyancy (kne)

2.1 Physical Context

The cosmic ray energy spectrum follows E?■■7 power law up to the "knee" at ~3■10■5 eV, where it steepens to E?■■■. The knee marks the maximum energy achievable by Galactic SNR shock acceleration for protons (Z=1). For heavy nuclei, the knee scales as Z ■ E_knee(proton) ■ the "knee composition model".

2.2 FUBiikne Equation

```
F_{\rm UBii,kne} = -F_{\rm rel} \cdot \frac{E_{\rm knee}}{E_{\rm GUT}} \cdot \frac{Z
\cdot e}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \ln\left(\frac{E_{\rm knee}}{E_{\rm
LEP}}\right)
```

where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

E_GUT = GUT energy scale = 1.6■10?5 J (~10■6 GeV)

Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
35.9
```

This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a* ***stationary point*** *of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
imes10^{-19})}{\ln(4.8\times10^{-4}/1.22\times10^{-19})} = 26 \times \frac{38.0}{35.9} =
27.5
```

The UQFF predicts the iron knee force is 27.5■ the proton knee force (vs 26■ in the pure rigidity model), a 5.8% enhancement from the quantum logarithmic correction.

---

3. Variant 9: WHIM Temperature Buoyancy (whim)

3.1 Physical Context

The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

```
F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

T_WHIM = WHIM temperature (K)

n_b = baryon number density (m?■)

s_T = Thomson cross-section = 6.652■10?■? m■

r_fil = filament radius (m)

T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

- Factor 1: thermal pressure k*B T*WHIM
- Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)
- Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K*, *n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m*, *T*virial = 107 K, Q_wave = 1.0:

```
F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

- M_halo = halo mass (kg)
- M_P = Planck mass = 2.176■10?8 kg
- d_c = 1.686 (critical overdensity)
- s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of M*P■ (the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10*■■ *M? = 1.989*■*104*■ *kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M**/M*gas ranges from ~1% in diffuse GMCs to ~30*■*50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
```

This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086*■*10*■*7 m, Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
(3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 =
1.72\times10^{22} \text{ N}
```

---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
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Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
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- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
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Validator: `BuoyancyProofVariants.py` *? All 17 FUBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER*{0:D3}" -f $n

---

Abstract

This paper presents five F*UBii buoyancy proof variants addressing quantum corrections to macroscopic astrophysical processes. Variant 7 (fermi) derives the UQFF buoyancy of Fermi-accelerated particles at astrophysical shock fronts. Variant 8 (kne) applies the framework to the cosmic ray knee at ~3■10■5 eV where the spectral index changes ■ the UQFF predicts this spectral break as a phase transition in the F*UBii landscape. Variant 9 (whim) addresses the Warm-Hot Intergalactic Medium at T ~ 105■107 K containing 40■50% of cosmic baryons. Variant 10 (ps) maps the Press-Schechter dark matter halo mass function to a UQFF buoyancy force landscape. Variant 11 (sfe) quantifies the UQFF buoyancy contribution to star formation efficiency suppression in molecular clouds. Together these form the quantum corrections series, where small-scale quantum physics drives large-scale structure.

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Key systems: Tycho SNR (v_shock ~ 4500 km/s), Centaurus A jet shocks, Cygnus A hotspots

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F_{\rm UBii,fermi} = F_{\rm rel} \cdot \frac{\beta_{\rm shock} \cdot E_p}{E_{\rm LEP}}
\cdot Q_{\rm wave} \cdot \left(\frac{v_{\rm shock}}{c}\right)^2
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where:

■_shock = shock compression ratio (typical 3■7 for strong shocks)

E_p = particle energy (J)

v_shock = shock velocity (m/s)

1.3 (v/c)■ Relativistic Correction

The Fermi buoyancy scales as (v*shock/c)*■ ■ *a relativistic correction that becomes important for v*shock > 0.1c. At non-relativistic shock speeds (v*shock/c ~ 0.01 for typical SNRs), the UQFF Fermi buoyancy is suppressed by (0.01)*■ *= 10?4 relative to the full E*p contribution.

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For Cen A hotspot: ■*shock = 4 (strong shock), E*p = 10?? J (10 GeV proton), v*shock = 0.5c, Q*wave = 1.0:

```
F_{\rm UBii,fermi}^{CenA} = 10^{-10} \times \frac{4 \times 10^{-9}}{1.22\times10^{-19}}
\times (0.5)^2 = 10^{-10} \times 3.28\times10^{10} \times 0.25 = 8.2\times10^{0} = 0.82
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The per-particle Fermi buoyancy force of 0.82 N per 10 GeV proton, scaled over the ~106■ protons in the hotspot, gives the collective Fermi acceleration pressure maintaining the Cen A jet head.

---

2. Variant 8: Cosmic Ray Knee Energy Buoyancy (kne)

2.1 Physical Context

The cosmic ray energy spectrum follows E?■■7 power law up to the "knee" at ~3■10■5 eV, where it steepens to E?■■■. The knee marks the maximum energy achievable by Galactic SNR shock acceleration for protons (Z=1). For heavy nuclei, the knee scales as Z ■ E_knee(proton) ■ the "knee composition model".

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```
F_{\rm UBii,kne} = -F_{\rm rel} \cdot \frac{E_{\rm knee}}{E_{\rm GUT}} \cdot \frac{Z
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```

where:

E_knee = knee energy (J) ■ 4.8■10?4 J (3■10■5 eV)

E_GUT = GUT energy scale = 1.6■10?5 J (~10■6 GeV)

Z = charge number of CR nucleus

e = 1.602■10?■? C

E_LEP = 1.22■10?■? J

The negative sign indicates spectral suppression ■ above the knee, CR buoyancy forces prevent further acceleration.

2.3 UQFF Knee as Phase Transition

The ln(E*knee/E*LEP) factor:

```
\ln\left(\frac{4.8\times10^{-4}}{1.22\times10^{-19}}\right) = \ln(3.93\times10^{15}) =
35.9
```

This logarithm attains its maximum precisely at E*knee for protons ■ a UQFF prediction that the knee is not an arbitrary cutoff but a **stationary point** of the F*UBii landscape where:

```
\frac{\partial F_{\rm UBii,kne}}{\partial \ln E} = 0 \quad \Rightarrow \quad E = E_{\rm
knee}
```

2.4 UQFF Knee Prediction: Proton vs Iron

For Z=1 (proton): E*knee = 3■10■5 eV = 4.8■10?4 J For Z=26 (iron): E*knee^Fe = 26 ■ 3■10■5 eV = 7.8■10■6 eV = 1.25■10?■ J

UQFF prediction for iron knee:

```
F_{\rm UBii,kne}^{Fe}/F_{\rm UBii,kne}^{p} = 26 \times \frac{\ln(1.25\times10^{-2}/1.22\t
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27.5
```

The UQFF predicts the iron knee force is 27.5■ the proton knee force (vs 26■ in the pure rigidity model), a 5.8% enhancement from the quantum logarithmic correction.

---

3. Variant 9: WHIM Temperature Buoyancy (whim)

3.1 Physical Context

The Warm-Hot Intergalactic Medium (WHIM) at z < 1 contains 40■50% of all baryons in the Universe ■ the "missing baryon problem" solution. It fills the cosmic web filaments at T ~ 105■107 K, traced by O VI (105.6 nm), O VII (21.6 ■), and soft X-ray emission. Its buoyancy against the cosmic gravitational potential maintains the filamentary structure of the Universe.

3.2 FUBiiwhim Equation

```
F_{\rm UBii,whim} = F_{\rm rel} \cdot \frac{k_B T_{\rm WHIM}}{E_{\rm LEP}} \cdot n_b
\sigma_T r_{\rm fil} \cdot Q_{\rm wave} \cdot \sqrt{\frac{T_{\rm WHIM}}{T_{\rm virial}}}
```

where:

- T_WHIM = WHIM temperature (K)
- n_b = baryon number density (m?■)
- s_T = Thomson cross-section = 6.652■10?■? m■
- r_fil = filament radius (m)
- T_virial = virial temperature of host structure (K)

3.3 T^(3/2) Scaling

The WHIM buoyancy scales as T*WHIM^(3/2)/(T*virial^(1/2)):

- Factor 1: thermal pressure k*B T*WHIM
- Factor 2: Thomson opacity depth n*b s*T r_fil (free electron count ■ cross-section)
- Factor 3: v(T*WHIM/T*virial) ■ buoyancy stability criterion (analogous to Schwarzschild stability for convection)

3.4 Example: Cosmic Web Filament (Sculptor Wall)

For a typical baryon-rich filament: T*WHIM = 106 K, n*b = 10 m?■, r*fil = 10 Mpc = 3.09■10■■ m, T*virial = 107 K, Q_wave = 1.0:

```
F_{\rm whim} = 10^{-10} \times \frac{1.381\times10^{-23} \times 10^6}{1.22\times10^{-19}}
\times 10 \times 6.652\times10^{-29} \times 3.09\times10^{23} \times \sqrt{0.1}
```

```
= 10^{-10} \times 1.132\times10^{2} \times 2.055\times10^{-4} \times 0.316 = 10^{-10}
\times 7.36\times10^{-3} = 7.4\times10^{-13} \text{ N}
```

This tiny force per unit volume, integrated over the filament volume V ~ (10 Mpc)■ ~ 3■107■ m■, gives F_total ~ 105? N ■ the UQFF buoyancy pressure holding the cosmic web filament against gravitational collapse.

---

4. Variant 10: Press-Schechter Halo Mass Buoyancy (ps)

4.1 Physical Context

The Press-Schechter (PS) mass function predicts the comoving number density of dark matter halos:

```
\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\rho_0}{M} \frac{\delta_c}{\sigma}
\left|\frac{d\ln\sigma}{d\ln M}\right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right)
```

where d_c = 1.686 is the critical overdensity for spherical collapse. The UQFF buoyancy force analog maps this statistical distribution to a physical force.

4.2 FUBiips Equation

```
F_{\rm UBii,ps} = -F_{\rm rel} \cdot \frac{M_{\rm halo}}{M_P^2} \cdot
\frac{\delta_c}{E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \left(-\frac{d\ln\sigma}{d\ln
M}\right)
```

where:

- M_halo = halo mass (kg)
- M_P = Planck mass = 2.176■10?8 kg
- d_c = 1.686 (critical overdensity)
- s = RMS density fluctuation

4.3 Planck Mass Normalization

The M*halo/M*P■ normalization is the UQFF quantum gravity anchor ■ halo masses are measured in units of MP■ *(the fundamental quantum gravity area unit). For a cluster-mass halo m*halo ~ 10■5 M?:

```
\frac{M_{\rm halo}}{M_P^2} = \frac{10^{15} \times
1.989\times10^{30}}{(2.176\times10^{-8})^2} =
\frac{1.989\times10^{45}}{4.74\times10^{-16}} = 4.2\times10^{60} \text{ m}^{-1} \cdot
\text{kg}
```

This enormous ratio reflects how macroscopic halo gravitational physics emerges from quantum Planck-scale foundations ■ a direct UQFF bridge between cosmological structure formation and quantum gravity.

4.4 Example: Milky Way Halo

For Milky Way: M*halo = 10■■ M? = 1.989■104■ kg, s(M*MW) ~ 0.5, dln s/dln M ~ -0.15, Q_wave = 1.0:

```
F_{\rm ps}^{MW} = -10^{-10} \times 4.2\times10^{57} \times
\frac{1.686}{1.22\times10^{-19}} \times 1.0 \times 0.15 = -10^{-10} \times
4.2\times10^{57} \times 1.38\times10^{19} \times 0.15 = -8.7\times10^{68} \text{ N}
```

---

5. Variant 11: Star Formation Efficiency Buoyancy (sfe)

5.1 Physical Context

Star formation efficiency e*SFE = M*/M*gas ranges from ~1% in diffuse GMCs to ~30■50% in dense molecular cloud cores. The UQFF buoyancy determines whether turbulence (e*SFE low) or gravity (e_SFE high) dominates in a given cloud.

5.2 FUBiisfe Equation

```
F_{\rm UBii,sfe} = F_{\rm rel} \cdot \frac{\varepsilon_{\rm SFE} \cdot M_{\rm gas} \cdot
c^2}{r_{\rm cloud}^2 \cdot E_{\rm LEP}} \cdot Q_{\rm wave} \cdot \sqrt{\varepsilon_{\rm
SFE}}
```

5.3 Rest-Mass Energy Term

The M_gas ■ c■ term is unusual in a cloud-physics context ■ it represents the UQFF's claim that star formation efficiency is ultimately limited by the rest-mass energy of the gas, mediated through the vacuum [SCm] manifold. The effective force is:

```
F_{\rm UBii,sfe} \sim \frac{\varepsilon^{3/2} M_{\rm gas} c^2}{r_{\rm cloud}^2}
```

This is the UQFF prediction that star formation is a quantum-gravitational process limited by the ratio of gas rest-mass energy to the cloud surface (Bekenstein-like area scaling).

5.4 Example: Orion A Giant Molecular Cloud

For Orion A GMC: e*SFE = 0.05, M*gas = 100 M? = 1.989■10■■ kg, r*cloud = 10 pc = 3.086■10■7 m, Q*wave = 1.0:

```
F_{\rm sfe}^{OrionA} = 10^{-10} \times \frac{0.05 \times 1.989\times10^{32} \times
(3\times10^8)^2}{(3.086\times10^{17})^2 \times 1.22\times10^{-19}} \times \sqrt{0.05}
```

```
= 10^{-10} \times \frac{0.05 \times 1.79\times10^{49}}{9.52\times10^{34} \times
1.22\times10^{-19}} \times 0.224 = 10^{-10} \times 7.68\times10^{31} \times 0.224 =
1.72\times10^{22} \text{ N}
```

---

6. Summary: Quantum Corrections Series

Variant	Physical Context	Key Formula	Characteristic Scale		
fermi	Fermi acceleration	■*shock* ■ *Ep* ■ (v/c)■	~0.8 N per 10 GeV proton		
kne	CR knee at 3■10■5 eV	E*knee/E*GUT ■ Z■e ■ ln(E/E_LEP)	Knee as F_UBii stationary pt		

Variant	Physical Context	Key Formula	Characteristic Scale		
whim	Cosmic baryon reservoir	k*BT* ■ *n*b s*T r*fil ■ v(T/T_vir)	~105? N per filament		
ps	PS halo mass function	M*halo*/*M*P■ ■ d_c ■	dln s/dln M		~1068 N (MW scale)
sfe	Molecular cloud SFR	e^(3/2) ■ M_gas c■/r■	~10■■ N (Orion A)		

Conclusions

Variants 7■11 demonstrate that the UQFF buoyancy framework provides quantitative predictions across the full range of quantum-to-cosmic scales:

- **fermi:** UQFF Fermi buoyancy provides the back-pressure that terminates DSA acceleration at the particle energy where F*UBii*fermi = F_confinement
- **kne:** CR knee is the stationary point of F*UBii*kne ■ a UQFF phase transition rather than a diffusive escape threshold
- **whim:** WHIM buoyancy F*UBii*whim ~ 105? N per filament maintains cosmic web structure against gravitational collapse
- **ps:** PS halo formation maps to Planck-mass-normalized UQFF collapse force ■ a quantum gravity bridge to large-scale structure
- **sfe:** Star formation efficiency is UQFF-limited by rest-mass energy Bekenstein-area scaling: F*UBii*sfe ? e^(3/2) M_gas c■/r■

Validator: `BuoyancyProofVariants.py` *? All 17 F_UBii variants operational ? | ? = 0.0005/day | [SSq] = 0.57*